University of Greater Manchester

COM4401 · APPLIED COMPUTER SCIENCE FUNDAMENTALS

Cyber-Security Awareness Report

Mini Project 4 · Team Report

A practical guide to staying safe online

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Declaration

Academic integrity and use of Generative AI (GAI)

We confirm that this group report is our own work. All sources have been acknowledged through appropriate in-text citations and references.

GAI tools used: Gemini, ChatGPT.

How they were used: To generate summary information, which we checked against authoritative sources, rewrote in our own words and understanding and referenced. Text was reviewed by the author.

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1. Introduction

Section lead: Chiamaka Eluwa **Peer reviewed by:** N/A

Most of us now bank, shop, study, work, and keep in touch online, so one compromised account can mean lost money, a stolen identity or access to everything linked to it. Many attacks do not break clever technology. They succeed because of ordinary habits, such as a weak or reused password, a message that is trusted too quickly, or a security step that is skipped.

This report is a practical guide for members of the public with no technical background. It explains five areas of everyday cyber-security awareness: password hygiene (Section 2), two-factor and multi-factor authentication (Section 3), phishing and social engineering (Section 4), safe use of devices (Section 5) and safe use of public Wi-Fi (Section 6). Section 7 brings the main findings together as a short list of actions that matter most.

1.1 Purpose and Importance

Cyber-security awareness means understanding the online threats you face and knowing the everyday habits that protect you, such as spotting suspicious messages, using strong passwords and keeping devices updated. It matters to individuals at home as much as to employees at work.

It is important because people's behaviour often decides whether an attack succeeds. Security software cannot stop someone from replying to a convincing scam or reusing a leaked password, and criminals know this. The stakes are also higher than ever, because banking, shopping, study and social life now run through linked online accounts. If one account is compromised, the damage can spread to the others.

1.2 Scope and Intended Audience

Who this report is for. The report is written for the general public, including people using the internet at home, at school or university, and at work. No technical knowledge is assumed. Each technical term is explained the first time it is used.

What it covers. Each of the five sections explains what the threat is, how the recommended protection works, what the evidence says and what readers can do. The report concentrates on personal accounts and devices rather than on the security systems of large organisations. Where a source is published by a recognised authority, such as the UK National Cyber Security Centre (NCSC), this is stated, so that readers can judge how far the advice can be trusted.

2. Password Hygiene

Section lead: Toby Mathew **Peer reviewed by:** Chiamaka Eluwa

2.1 Understanding Password Hygiene

Password hygiene is practicing a good set of password habits to ensure that your online accounts are kept safe from black hat hackers. There are many practices when it comes to password hygiene. These include creating long unique passwords (roughly 12-16 characters long. includes a mix of letters, numbers and symbols), never reuse the same password, using multi factor authentication, do not under any circumstances share details with anyone (not even family, friends or coworkers), and watch out for phishing by not trusting strange or fake emails. Also make sure it is not easily discoverable. This includes using passwords based off personal data or preferences.

There are many ways in which a password can be compromised. These can be things such as credentials of stuffing, phishing, data breach, dictionary attacks, malware, shoulder surfing, and password reuse. Credential stuffing is when an attacker finds a username/password pair and tries them on multiple sites to try and login on them. Dictionary attacks are when hackers use automated guessing via wordlists of common passwords, names, dates, and patterns. Phishing is when criminals pretend to be trustworthy to extract sensitive information. Shoulder surfing is when a criminal peers over someone's shoulder to see information; it is commonly used with PIN number for bank cards. Password reuse is when a user uses the same password repeatedly. Long and unique passwords improve security as they make a password harder and more confusing to remember. Not reusing a password causes there to be more passwords to remember which in turn will confuse the hacker. Multi-factor authentication adds another layer of protection to act as a barrier.

2.2 Password Managers and Password Reuse

Another great way to increase password hygiene is to use password managers. password manager both creates random passwords and stores them for the users' later use. Password managers work by setting a master password which the manager uses. Logins are then stored in encrypted vaults using methods like XChaCha20 or AES-256. The vault syncs across all the user's devices, and an app or extension automatically fills in the required information when you visit a matching site. When an account is created, it generates a long random password and saves it automatically.

The benefits of password managers are that their passwords are unique without using memory, passwords are both strong and random, reduce phishing due to only auto filling on exact domains, flags when passwords are weak or reused, convenient for fast logins and multidevice syncing, and it provides emergency recovery options. However, there are risks such as there being a single point of failure, no access if the master password is forgotten, and susceptible to malware.

If the manager is breached, then it puts other accounts at risk. This is cause when the manager is breached, everyone's credentials get exposed, bots then try those username/password pairs everywhere until they work. When they get the email, they can use the forgotten password feature to reset every other account. From then on, they can use your account to commit crimes like fraud, stealing data, impersonating contacts, and locking you out. A real-life example of the consequences of compromised credentials is the colonial pipeline in 2021. Colonial pipeline is an American oil company. They suffered a ransomware cyberattack that compromised all the equipment of the pipeline managed by computers. They ended up

paying the hackers 75 bitcoins or roughly £3.2 million at the time. It was the largest cyberattack on an oil company bin the history of the USA

2.3 Evidence and Recommended Practice

[TO BE COMPLETED – Toby Mathew: Support the Do and Don't list in Table 1 with cited evidence.]

Do	Don’t
Use long and unique passwords	Reuse the same password
Use MFA	Share your details
Use password managers	Base the password off personal information

Table 1: Password do’s and don’ts (Source: NCSC, NIST Special Publication, OWASP, CISA, FIDO Alliance)

3. Two-Factor and Multi-Factor Authentication

Section lead: Chiamaka Eluwa **Peer reviewed by:** N/A

3.1 What is 2FA/MFA?

Authentication is the process of proving that you are who you say you are when you use an online service. A password is the most common proof, but a password is only one piece of evidence and, as Section 2 shows, it can be guessed, stolen or reused. **Two-factor authentication (2FA)** asks for a second, different kind of proof before access is allowed. **Multi-factor authentication (MFA)** is the wider term for two or more. The NCSC uses the name 2-step verification (2SV) for the same idea and describes it as a way of double-checking that you are who you claim to be (NCSC, 2022).

The proofs, called factors, fall into three categories (Temoshok et al., 2025):

- **Something you know**, such as a password or PIN.
- **Something you have**, such as a phone that receives a code, an authenticator app or a physical security key.
- **Something you are**, such as a fingerprint or a face scan.

The factors must come from different categories. A password followed by a security question is two steps, but both are things you know, and the NCSC notes that this does not give the same protection (NCSC, 2022).

The extra factor matters because a stolen password is no longer enough. If a criminal obtains your password through a phishing email, a data breach or guesswork, they still cannot sign in without also controlling your second factor. The NCSC recommends turning on 2SV for important accounts and especially for email, because access to an inbox can be used to reset the passwords of your other accounts (NCSC, 2022).

3.2 Comparing Authentication Methods

Not every second factor offers the same protection. Three common choices are SMS codes, authenticator apps and hardware security keys. Table 2 compares their strengths and limitations.

Method	How it works	Strengths	Limitations
SMS code	A one-time code is sent by text message (or a voice call) to the user's phone number and typed into the website.	Offered by most services by default, so it is easy to start using. Still a big improvement on a password alone (NCSC, 2022).	CISA describes SMS and voice codes as vulnerable to phishing, SS7 and SIM-swap attacks and a “last resort” option (CISA, 2022). Phone carriers' checks before a SIM swap can be defeated (Lee et al., 2020). Needs mobile signal.

Method	How it works	Strengths	Limitations
Authenticator app	An app on the phone creates a short-lived code, or asks the user to approve a sign-in.	The NCSC describes apps as the main alternative to text messages. The code is created on the device, so no mobile signal or waiting for a text is needed (NCSC, 2022).	A fake login page can still trick users into typing the code. Approval prompts without number matching are open to "push bombing", where users are flooded with requests (CISA, 2022). A lost phone needs backup codes (NCSC, 2022).
Hardware / security key	A small physical device, plugged in or tapped on the phone or computer, proves the user holds it using the FIDO/WebAuthn standard.	FIDO/WebAuthn is the only phishing-resistant option CISA describes as widely available and it calls it the "gold standard" (CISA, 2022).	Must be bought and carried, and is offered by only some services (NCSC, 2022). A backup plan is needed if the key is lost (NCSC, 2022).

Table 2: Comparison of authentication methods (Source: Source: CISA, 2022; Lee et al., 2020; NCSC, 2022)

3.3 Evidence and Recommended Practice

What the evidence shows. Google studied more than 350,000 real attempts to hijack accounts. Challenges based on knowledge, such as security questions, prevented as few as 10% of phishing-based attempts and 73% of automated attempts. Challenges based on a device the user holds performed best, blocking over 94% of phishing-based attempts and 100% of automated ones (Doerfler et al., 2019).

Weaknesses of different methods. Text messages depend on the phone network. A study of five prepaid US phone carriers found that all five used checks that attackers could easily defeat when asking for a number to be moved to a new SIM card. The same study found 17 websites where a SIM swap alone, without the password, was enough to take over an account (Lee et al., 2020).

What authorities recommend. The NCSC, writing for the public, stresses that any 2SV is better than none and presents authenticator apps as an advantage over text messages (NCSC, 2022). The US Cybersecurity and Infrastructure Security Agency (CISA) goes further, calling FIDO/WebAuthn security keys the strongest option and SMS a last resort (CISA, 2022).

What readers should do:

- Turn on 2FA/MFA first for email, then for banking, social media and shopping accounts.
- Choose the strongest method the service offers. Use a text message only if nothing better is available.
- Save backup codes somewhere safe, so a lost or flat phone does not lock you out (NCSC, 2022).
- Do not approve a sign-in request you did not start (CISA, 2022).

4. Phishing and Social Engineering

Section lead: Enrique Manzano Ayala **Peer reviewed by:** Chiamaka Eluwa

4.1 Understanding Phishing

What is it known as phishing?

Phishing is a type of cybercrime that involves attacker impersonating either individuals that are known and trusted to the individual or organizations or companies the individual trusts and that trick them in to revealing their own sensitive information or even giving them access to their devices systems , at difference of many other types of cyber-attacks phishing isn't one that targets technical defenses and tries to break through them phishing goes in more deep as this one turns to exploit the human trust such as habit and emotions. As communication through virtual networks and apps become part of our day-to-day life , in work and our day to day , phishing has started to become one of the most common and worldwide threats online not only affecting individuals but also affecting bigger companies as this happens by this cyber attackers impersonating someone that's very close to the individual an great example is when you receive an email from your bank the original email being : *generalexampleinfobank@genericbank.co.uk* and the cyber attacker using an email like: *generalexampleimfobank@gennericbank.co.uk* at simple sight you can't really tell the difference as when the email is sent the name it comes under is Generic Bank , and the attackers could just be : Generic Banc , as by simple things like that people loses their data , money accounts .

Smishing and Vishing

Both of them target a specific user but they have different approaches , smishing happens via text message but in other hand vishing happens by the approach of an phone call , either of them can be mass produced or specifically target , and the choice of the method of approach changes the way the victim reacts , and short text message is easy as it makes you unconsciously tap it , in which smishing messages happen to be a one line prompt with an URL link which most of the times happen to be the usual , missed parcel and that they need to pay for the re delivery. A phone call is more direct and effective as it allows the attacker to pressure the victim responds to doubts and keeps the victim from stopping to think and analyze what it is being asked and called about, that's why vishing is so effective when it comes that the caller poses as a bank fraud team.

Spear phishing differs on its targets as , is a more specialized attack, as an normal phishing email , goes to thousands of people , and they hope a few respond to it , while spear phishing is aimed to one specific individual or group of this ones , using their real details such as their real details as their job , work colleagues , recent activity to make the message more believable and coherent for the person they targeting , it can arrive by an email , text or phone so a smishing message that includes your actual bank and a recent purchase also counts as spear phishing.

4.2 Why Phishing Works

Most of the phishing attacks are target to individuals that looked from a different point they are marked as vulnerable in a certain way and it makes more sense that someone has an real monetary situation receives an message or an email that tells him that he can be helped or supply more founds or reduced from their total ,and the urgency and fear comes when they receive a message that comes in with their details and the bank name saying that there has been and no notified use of their cards and taken out of money which sometimes exceeds that mount that this person has on their account which leads them to no doubt to

touching the URL that tells the they can get It fixed and back to them If they follow certain steps and they make an safe deposit to know where to "send" the refund , authority could be by an cyber attacker impersonating an company superior saying that changes have to be made to the company and that now they to send the actions and everything to different places when in reality that person that the attacker absolutely has no clue about this things neither the company , another type of manipulation that isn't really recured to is the familiar in problem , an distant familiar is the excuse most of the times in some extreme cases is the kidnapped familiar , but they don't happen that often as this one's usually work when the approach is to elderly people and they all have effect as they target the weakness of the individuals which are the ones that can make an mayor harm on them and that's why it works .

4.3 Common Warning Signs

Warning sign	Why it may be suspicious	What the user should do
A spelling mistake in the name or title or message itself.	If it comes from the pace it says I is which sometimes is from a professional background mistakes in there and things out of place are an absolute no and most of them emails have their own @somthing.somehting.som there should be the ones as they part of a line of connection of business anything that's in the name of that company that isn't like that is no .	Be very observant when it comes to the details and if they receive any message of email that looks very convincing, they could do the best strategy ever: search and compare with an email or message from that place that you can recall being the right one and sent from the right place to
Addressing of the individual	If the name is on full CAPITAL letters as if is a professional strict message the message should be written properly addressing and using the capital letter only if required not for the whole message	IF THE MESSAGE STARTS ADDRESSSING YOU BY THE WRONG NAME OR ALL IN CAPITALS AND IS SENT IN THE NAME OF A COMPANY OR A BANK don't even give them the benefit of opening it as anything that's from a professional or real place shouldn't be in full capital letters and the name they address you on has to be the. Right one.
Phone number used to send the message or made the call start by 073 or 078	Companies contact and phone numbers will never start as if they were an personal phone number as when it comes to most numbers of companies and places start by the numbers 0161 o 0300 which are companies numbers	If the number starts by an 07 or +44 7 that's and circling number that you or someone else could use So if it starts by that is mostly guaranteed and scam number so do please check and if the number has the wrong start put it as scam or simply don't touch.

Table 3: Common phishing warning signs (Source: [citation])

4.4 Responding to Phishing

If you suspect that a message is a scam, avoid responding to it, clicking on any links or opening attachments . GOV.UK recommends that people do not share personal information or interact with suspicious messages. Instead, phishing emails should be reported to report@phishing.gov.uk, while suspicious text messages ca n be forwarded to 7726 free of charge. Scam calls can also be reported by texting “Call” followed by the tel ephone number to 7726. The National Cyber Security Centre (NCSC)examines reported emails and linked w ebsites to identify potential threats and take appropriate action.

5. Safe Use of Devices

Section lead: Fatimah Bibi **Peer reviewed by:** Chiamaka Eluwa

5.1 Updates, Vulnerabilities and Patches

Every phone, laptop and app is built from millions of lines of code, and mistakes in that code are inevitable. A mistake that an attacker can abuse is called a *vulnerability* (NIST, 2024). The tool or technique used to take advantage of it is an *exploit* (NIST. When the manufacturer fixes the mistake, the fix is called a *patch*, and it reaches users as a software update (NCSC, 2026). The three are locked in a race: once a vulnerability becomes known, criminals try to build exploits while manufacturers try to release patches. An update is the only way a patch reaches your device, so putting it off leaves the door open even though the lock has already been fixed.

The WannaCry attack shows what this costs. In May 2017, WannaCry ransomware (malicious software that locks files and demands payment) spread through the NHS in England. NHS Digital had issued critical alerts in March and April 2017 telling organisations to patch their systems, yet every NHS organisation that was infected shared the same vulnerability and could have protected itself with relatively simple action (NAO, 2017). The warning and the fix both existed; the damage came from the fix not being applied.

For individuals, the National Cyber Security Centre (NCSC) advises installing the latest operating system and app updates and switching on automatic updates where possible. It adds that updates contain security fixes as well as new features, and that if you are busy you should find time later the same day rather than ignoring them (NCSC, 2026). You also do not need the newest model, only a device that still receives updates. Apps that no longer receive updates should be replaced with supported alternatives, because an unsupported app will never be patched (NCSC, 2026).

5.2 Protecting Devices and Data

Screen locks. A screen lock is the first barrier if a device is lost, stolen or left unattended. The NCSC recommends setting a passcode that locks the device automatically after a period of inactivity, which makes it harder for an attacker to misuse a device that is lost, stolen or left unlocked (NCSC, 2021).

Device encryption. Encryption scrambles the data stored on a device so that it is unreadable without the correct key, which on most devices is derived from your passcode. Even if a thief removes the storage, the files stay unreadable. The NCSC recommends enabling device and storage encryption where possible (NCSC, 2021). This is also why a strong passcode matters: encryption is only as strong as the secret that unlocks it.

Locating, locking and erasing. Manufacturers provide websites where you can log in from another device, locate a lost device, and lock or erase it remotely. Sometimes you must enable this feature before the device is lost (NCSC, 2021) so it is worth switching on now. Before selling or recycling a device, copy across what you want to keep and then perform a full factory reset (NCSC, 2020).

Security software. Antivirus software stops computer viruses from harming a device, while a firewall blocks unwanted connections, usually from the internet, from reaching it (NCSC, 2026). Neither is perfect, because security software can only act on threats it can recognise. It works best alongside updates rather than replacing them.

Installing software safely. Apps from a device's built-in store are vetted before release, so they can be checked for security issues (NCSC, 2023). Deleting apps you no longer use reduces the number of

programs that could contain a flaw, and it also saves storage (NCSC, 2019). Installing apps from outside the official

5.3 Other Device Risks

Apps from unofficial sources. Installing apps from outside the official store, known as "sideloading", bypasses the vetting described above. Malware (software designed to damage a device or steal data) can be hidden inside an app that looks genuine, and nobody has checked it before it reaches you. The simplest protection is to install apps only from the built-in store (NCSC, 2017).

Unknown USB devices. In a field experiment, researchers dropped 297 USB drives around a university campus. Almost half were picked up and had at least one file opened (Tischer *et al.*, 2016). Plugging in a found drive can run malicious software, so unknown USB devices should be handed to the owner or to IT support rather than plugged in. People should not plug in USB devices if they do not know what it contains.

Public charging points. The US Federal Communications Commission (FCC) warns that malware installed through a compromised USB port can lock a device or export personal data and passwords. This tactic is called "juice jacking". It also notes that criminals have in some cases left cables plugged into charging stations (FCC, 2023). The advice that they give is to use an AC power outlet rather than a public USB port, carry your own charger and cable, and consider a charging-only cable that blocks data transfer (FCC, 2023). It is worth being honest about how big the risk is, though. Security experts quoted by AARP (2023) said they had not seen the threat at public charging stations, and the FBI described its own warning as a general reminder rather than a response to a specific threat. The risk is technically possible rather than widely documented, but the precautions cost almost nothing, so they remain sensible.

6. Safe Use of Public Wi-Fi

Section lead: Enrique Manzano Ayala **Peer reviewed by:** [Name]

6.1 Understanding the Risks

GUIDANCE *— delete this box before submission*

Explain open or unsecured networks, eavesdropping, man-in-the-middle attacks and “evil twin” hotspots, and how each could affect someone on a public network.

[TO BE WRITTEN – Enrique Manzano Ayala: 6.1 Explain open networks, eavesdropping, man-in-the-middle attacks and evil twin hotspots.] Don’t forget to delete the guidance box after.

6.2 HTTPS and VPNs

GUIDANCE *— delete this box before submission*

Explain what HTTPS and a VPN each protect, and the difference between them. Avoid saying either one simply “makes you safe” — explain its **limitations** too.

[TO BE WRITTEN – Enrique Manzano Ayala: 6.2 Explain what HTTPS and a VPN each protect, and their limitations. Then complete Table 4.] Don’t forget to delete the guidance box after.

	HTTPS	VPN
What does it do?		
What does it protect?		
What does it not protect against?		

Table 4: HTTPS and VPNs compared (Source: [citation])

6.3 Recommended Practice

GUIDANCE *— delete this box before submission*

Research what recognised authorities currently recommend. Check the **publication dates** of your sources — advice on public Wi-Fi has changed as technology has developed. Where older and newer advice differ, explain how and why.

[TO BE WRITTEN – Enrique Manzano Ayala: 6.3 Give current recommended practice. Check publication dates and explain where older and newer advice differ.] Don’t forget to delete the guidance box after.

7. Conclusion

Section lead: Chiamaka Eluwa **Peer reviewed by:** N/A

7.1 Summary of Findings

Across the five areas, one of them stands out: attackers usually look for the easiest way in, and the easiest way in is often a person rather than a piece of software. The second theme is that no single measure is enough. A strong password can still be phished, a password manager is a single point of failure if it is breached, and even a second factor varies in strength. Protection comes from layers, and the choice of layer matters: device-based checks blocked far more phishing-based attempts than knowledge-based ones (Doerfler et al., 2019).

If readers remember only five things, they should be these:

- **Use long, unique passwords** and let a password manager remember them (Section 2).
- **Turn on 2FA/MFA**, starting with email, and use the strongest method available. Any second step is better than none (NCSC, 2022) (Section 3).
- **Pause before acting on an unexpected message or call**. Do not click, reply or pay, check through an official channel, and report it (Section 4).
- [Safe use of devices – Fatimah Bibi to add one action from Section 5]
- [Public Wi-Fi – Enrique Manzano Ayala to add one action from Section 6]

Small, regular habits such as these cost little and make an attacker's job much harder.

8. References

Cybersecurity and Infrastructure Security Agency (CISA) (2022) *Implementing phishing-resistant MFA*. Available at: https://cisa.gov/sites/default/files/publications/fact-sheet-implementing-phishing-resistant-mfa-508c.pdf (Accessed: 22 October 2026).

Doerfler, P., Marincenko, M., Ranieri, J., Jiang, Y., Moscicki, A., McCoy, D. and Thomas, K. (2019) 'Evaluating login challenges as a defense against account takeover', *Proceedings of The World Wide Web Conference (WWW '19)*. New York: Association for Computing Machinery. Available at: https://research.google/pubs/evaluating-login-challenges-as-a-defense-against-account-takeover/ (Accessed: 22 October 2026).

Lee, K., Kaiser, B., Mayer, J. and Narayanan, A. (2020) 'An empirical study of wireless carrier authentication for SIM swaps', *Proceedings of the Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020)*. Berkeley, CA: USENIX Association, pp. 61–79. Available at: https://www.usenix.org/conference/soups2020/presentation/lee (Accessed: 22 October 2026).

National Cyber Security Centre (NCSC) (2022) *Setting up 2-Step Verification (2SV)*. Available at: https://www.ncsc.gov.uk/guidance/setting-2-step-verification-2sv (Accessed: 22 October 2026).

Temoshok, D., Fenton, J., Choong, Y.-Y., Lefkovitz, N., Regenscheid, A., Galluzzo, R. and Richer, J. (2025) *Digital identity guidelines: authentication and authenticator management*. NIST Special Publication 800-63B-4. Gaithersburg, MD: National Institute of Standards and Technology.

Colonial Pipeline Ransomware Attack:

Colonial Pipeline ransomware attack - Wikipedia

NIST Special Publication:

https://csrc.nist.gov/pubs/sp/800/63/b/4/final

NCSC:

https://www.ncsc.gov.uk/collection/top-tips-for-staying-secure-online/password-managers

OWASP:

https://cheatsheetseries.owasp.org/cheatsheets/Credential_Stuffing_Prevention_Cheat_Sheet.html

FIDO Alliance:

https://www.passkeycentral.org/introduction-to-passkeys/how-passkeys-work

[Toby, Enrique and Fatimah: add your references here in alphabetical order (add the Colonial Pipeline and GOV.UK/NCSC sources), then remove this line.]

Appendix A: Team Contribution Record

Team member	Sections written	Approx. words	Contribution to intro / conclusion	Sections peer reviewed
Toby Mathew	Section 1.1, 2	574	Wrote part of Section 1.1	
Chiamaka Eluwa	Sections 1, 3, 7.1	1390	Wrote the introduction, Section 1.2 (Scope and Intended Audience), refined and completed Section 1.1, and wrote Section 7 (conclusion).	
Enrique Manzano Ayala	Section 4		N/A	
Fatimah Bibi	Section 5		N/A	

Appendix B: Individual Source Log

Each student: at least **5 credible sources**, including at least **1 peer-reviewed** and **2 authoritative professional or government** sources.

Student	Source (Harvard reference)	Type	Used in section
Chiamaka Eluwa	Doerfler, P., Marincenko, M., Ranieri, J., Jiang, Y., Moscicki, A., McCoy, D. and Thomas, K. (2019) 'Evaluating login challenges as a defense against account takeover', Proceedings of The World Wide Web Conference (WWW '19). New York: Association for Computing Machinery. Available at: https://research.google/pubs/evaluating-login-challenges-as-a-defense-against-account-takeover/ (Accessed: 22 October 2026).	P	3.3, 7.1
Chiamaka Eluwa	Lee, K., Kaiser, B., Mayer, J. and Narayanan, A. (2020) 'An empirical study of wireless carrier authentication for SIM swaps', Proceedings of the Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020). Berkeley, CA: USENIX Association, pp. 61–79. Available at: https://www.usenix.org/conference/soups2020/presentation/lee (Accessed: 22 October 2026).	P	3.2, 3.3
Chiamaka Eluwa	Cybersecurity and Infrastructure Security Agency (CISA) (2022) Implementing phishing-resistant MFA. Available at: https://cisa.gov/sites/default/files/publications/fact-sheet-implementing-phishing-resistant-mfa-508c.pdf (Accessed: 22 October 2026).	A	3.2, 3.3, 7.1
Chiamaka Eluwa	National Cyber Security Centre (NCSC) (2022) Setting up 2-Step Verification (2SV). Available at: https://www.ncsc.gov.uk/guidance/setting-2-step-verification-2sv (Accessed: 22 October 2026).	A	3.1, 3.2, 3.3, 7.1
Chiamaka Eluwa	Temoshok, D., Fenton, J., Choong, Y.-Y., Lefkovitz, N., Regenscheid, A., Galluzzo, R. and Richer, J. (2025) Digital identity guidelines: authentication and authenticator management. NIST Special Publication 800-63B-4. Gaithersburg, MD: National Institute of Standards and Technology.	A	3.1

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Type: P = peer-reviewed academic · A = authoritative professional / government · O = other credible source